



UNIVERSITY OF CENTRAL FLORIDA

Final Project Report: Digital Twins for Natural Disaster Operations

NetProtect: Digital Twin for Resilient Communication Networks

Document Summary: The Lahaina wildfire of 2023 exposed the serious danger of faulty communication during natural disaster operations which can be optimized through forming a communications network around a theorized Digital Twin.

Connor Hatfield, Dallas Nash, Alicia Thoney • 04/23/2025 • IDS6938 CMB-25Spring 00218

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1. Introduction

Wildfire-driven disasters expose a harsh reality: when communications infrastructure fails, every other facet of response falters. The 2023 Lahaina, Maui wildfires illustrated this vividly: cellular towers went dark, emergency alerts never reached residents, and first-responder radio channels operated in isolation, leaving public officials largely “in the dark” during the critical first hours (Press, 2024; AP News, 2024; Guardian News and Media, 2024). Over 100 lives, 2,153 structures, and \$5.5 billion were lost to the fire ravaging the community (Hedayati, 2024). After-action reports highlighted three systemic shortcomings: fragmented data flows, minimal redundancy across networks, and the absence of a unifying architecture for real-time decision support.

In response, we have undertaken Net-Protect: a research-driven initiative to design and prototype a digital twin of disaster-response communications networks for Lahaina and, by extension, similar at-risk communities. A digital twin (DT) is a dynamic virtual representation of physical assets and processes that remains continuously synchronized with its real-world counterpart (Digital Twin Consortium, 2024). While DTs are widely discussed for smart-city or industrial applications, our focus is deliberately narrow, yet mission-critical: to ensure that first responders and public officials maintain uninterrupted, situationally aware communication before, during, and after a natural hazard.

The intellectual and practical motivation behind Net-Protect is two-fold. First, we aim to shift disaster communications from a *reactive* stance—patching outages once they occur—to a *predictive* and *prescriptive* posture in which failures are forecast and mitigated in advance. Second, we seek to provide a transparent, research-based framework that agencies can replicate or adapt, thereby advancing the broader body of knowledge on digital twins for emergency management. Our approach aligns specific capabilities from the Digital Twin Consortium’s Capabilities Periodic Table—such as Data Acquisition & Ingestion, Simulation, Prediction, and Prescriptive Recommendations—with the practical needs of redundancy, interoperability, and real-time analytics identified in Lahaina’s after-action findings.

This report documents our journey from problem framing through methodology design, preliminary results, and an implementation roadmap. By detailing both the successes and the challenges encountered, we intend to furnish the emergency-management research community with a blueprint that bridges academic inquiry and operational utility. Ultimately, Net-Protect demonstrates that a focused, data-driven digital twin can transform communications resilience, turning lessons learned from Lahaina into actionable foresight for future disasters.

2. Background

2.1 The Lahaina Wildfire in Context

On 8 August 2023, Hurricane Dora’s peripheral winds—gusting to 50 mph while relative humidity dropped below 30 percent—collided with months-long drought on Maui’s leeward coast. The National Weather Service issued Red-Flag Warnings three days earlier, but no surge staffing or pre-positioning of resources took place (WFCA, 2024). By mid-afternoon a rekindled vegetation fire east of Lahainaluna Road “spotted” far ahead of containment lines and, within minutes, became an urban conflagration that destroyed 2,170 structures and claimed 101 lives (WFCA, 2024).

2.2 Communications Landscape Before the Fire

Table #1: 2023 Communication Assessment

Layer	Pre-incident status	Latent vulnerabilities
Public alerts (WEA texts, outdoor sirens)	County relied primarily on Wireless Emergency Alerts (WEA); five outdoor warning sirens installed in Lahaina.	Four of the five sirens were tagged “Beyond Repair” and offline as of July 2023; no policy for wildfire siren activation.
Commercial telecom	21 cell sites served West Maui (FCC DIRS-Lite baseline).	Above-ground back-haul and commercial generators offered limited backup, no hardened battery or satellite fail-over.
Public-safety land-mobile radio (LMR)	700/800 MHz trunked system with repeaters on Lāna‘i; five console positions in Central Dispatch.	Normal staffing of three dispatchers; channels already busy during multi-incident days.

Dispatch & 9-1-1 center	Authorized for 39 personnel, but only 12 positions filled; three on duty per shift.	Minimal surge capacity; only one overflow console for major events.
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2.3 Communications Failures (8 – 9 August 2023)

1. Public warning did not penetrate.

- a. Interviews and survey data show 87 percent of residents received no WEA text during the crisis (FSRI Finding 73).
- b. County officials declined to activate the siren network; with just a single operable unit they feared confusing residents unfamiliar with its wildfire meaning.

2. Commercial networks collapsed.

- a. As poles burned and generators failed, coverage shrank rapidly; by 9 August the FCC tallied all 21 Lahaina cell sites out of service (FCC, 2023) [FCC Documents](#).
- b. Private water-tanker contractors—who coordinate solely by mobile phone—lost contact once the network failed, forcing face-to-face dispatching.

3. Radio systems survived but became congested.

- a. Tactical talk-groups peaked at 10–12 transmissions · min⁻¹, forcing responders to queue for airtime.
- b. Wind noise and missing earpieces degraded message intelligibility (FSRI Finding 74).

4. Dispatch overload.

- a. 9-1-1 call load surged to 4,529 contacts, nearly triple the daily average, while only three dispatchers and five consoles were available.

5. Lack of unified command & improvised field alerts.

- a. The county’s mobile command vehicle was never deployed; incident commanders worked from moving vehicles, fragmenting situational awareness.
- b. The Maui Police Department resorted to vehicle public-address systems for evacuation orders, but wind, smoke, and elders’ hearing loss limited audibility.

2.4 Where We Are Now (April 2025)

Table #2: After Action 2023 Fire Management

Domain	Post-fire status	Remaining gaps
Siren network	HI-EMA adopted an all-hazards activation policy; four Lahaina sirens remain offline (MEMA AAR Key Finding 4) Maui County .	Stable funding stream and maintenance crews still unresolved; wildfire-specific public-education campaign pending.
Dispatch & radio	Study underway to expand console count and create a cross-trained surge pool (FSRI Findings 32-34).	No surge cadre hired; congestion-mitigation hardware (extra channels, incident repeaters) unfunded.
Public alerting	County evaluating multi-channel program that pairs WEA with FM/AM broadcasts, NOAA radio, and neighborhood liaison teams (MEMA AAR) Maui County .	Dependence on commercial carriers persists; no satellite or HF backup path in place.
Inter-agency comms	Draft plan for pre-staging a mobile Incident Command Post on red-flag days.	Memoranda of understanding and radio-equipment cache funding still in negotiation.

2.5 Why Communications Must Be the First Target for Change

Every operational bottleneck during the Lahaina disaster—from delayed evacuations to uncoordinated resource deployment—traced back to failures in information flow. Unlike terrain or weather, communications resiliency is an engineering challenge we can directly address. By modelling load, redundancy, and failure modes in a dedicated digital twin, we intend to transform the hard-won lessons above into predictive, prescriptive capability—ensuring the next red-flag day does not devolve into another communications blackout.

3. Problem Statement & Requests for Action

3.1 Why Change Is Urgent

The 2023 Lahaina fire proved that a single-failure communications chain can turn a wildfire into a mass-casualty event. Outdoor sirens were inoperable, Wireless Emergency Alerts reached few residents, every commercial cell site failed within 18 hours, and radio/dispatch

channels became saturated beyond usable limits (FSRI, 2024; FCC, 2023; WFCA, 2024; MEMA, 2025). These gaps delayed evacuations, fragmented command, and cost lives.

3.2 Statement of Need

We must convert West Maui’s disaster-communications network from a reactive, single-point system into a predictive, redundant, and data-driven architecture that guarantees uninterrupted information flow for responders and the public throughout any natural-hazard event.

3.3 Change Objectives (Requests for Action)

Table #3: Request for Action

#	What we must build	Why it matters
1	Single-Source-of-Truth Digital Twin that ingests live telemetry from towers, sirens, power feeds, weather, and social channels.	Eliminates siloed data and gives all agencies the exact same real-time picture.
2	Simulation + Predictive Analytics Engine inside the twin that stress-tests the network, forecasts outages, and generates prescriptive recommendations (e.g., “deploy COW,” “invoke GETS”).	Moves the system from reactive to predictive decision-making.
3	Multi-Channel Alert & Decision Dashboard —a web/mobile interface and API that push twin insights to dispatch, EOC, and field units, and pull status updates back in.	Ensures responders receive actionable information without changing their existing procedures.
4	Continuous-Learning Loop that feeds drill logs, synthetic scenarios, and post-incident data back into the twin to refine models and alert threshold.	Keeps the platform accurate as hazards, infrastructure, and user needs evolve.

4. Methodology

4.1 Expert-Centered Inquiry

Our fact-finding revolved around two subject-matter experts:

- **Christopher Vaughan – Incident-Command Veteran**

- **Engagement.** Three scheduled video sessions plus ad-hoc emails.
- **Focus areas.**
 - Current communications workflow and pain points.
 - First-hand lessons from the Lahaina fire.
 - “Must-have” features for a communications-oriented digital twin.
 - Recommended data sources, jargon, and prior student work on this topic.
- **Dr. Soheil Sabri – Digital-Twin Scholar**
 - **Engagement.** On-call guidance.
 - **Focus areas.**
 - Validating our scope and capability mapping.
 - Ensuring alignment with accepted digital-twin terminology and practice.
 - Reality-checking feasibility against known DT implementations.

Each interview followed a semi-structured script (open questions → clarifying probes → verbal recap). Notes were transcribed immediately and circulated for confirmation.

4.2 Iterative Question–Refine Loop

1. **Ask.** Capture qualitative insights from the experts.
2. **Reflect.** Distill conversations into provisional requirements.
3. **Verify.** Return the draft requirements to the experts for quick corrections.
4. **Refine.** Adjust the next interview’s focus based on verified gaps.
5. **Document.** Maintain a living “needs matrix” that matured into the four Change Objectives in Section 3.3.

4.3 Outcome of the Method

This tight, interview-driven cycle kept our effort grounded in operational reality: responders do **not** need new procedures; they need a single, trustworthy data picture that predicts failures before they occur. The objectives we formulated—and the roadmap that follows—trace directly to that expert-validated mandate. A mixed methods approach was used for research utilizing quantitative and qualitative data for communication, geography, and wildfire data. TOGAF, DT Consortium, and previously published wildfire management DT frameworks were

used to formulate the architecture vision subsequently following into the business architecture, information systems architecture, and technology architecture. The result of the frameworks and literature analysis resulted in a roadmap to achieve the goal of implementing the DT in a hypothetical three years.

5. Results

5.1 Vision Architecture

5.1.1 Maturity Assessment

Table 1 details the current state of the communication network used by emergency responders in Lahaina. The goal of this maturity assessment is to create a baseline for what needs to be changed or modified to reach the target digital twin architecture. The result demonstrated that the basic systems in place lack communication network interoperability, IoT real-time sensors, and reliable infrastructure.

Table #4: Maturity Assessment

Maturity Assessment criteria	Basic	Developing	Advanced	Justification(s)
Data-driven decision making	X			There was a lack of data integration and predictive analytics in the emergency response efforts. Sources say that this lack of real-time data hindered proactive measures and timely decision making
Integration of technology	X			The existing emergency communication systems in Lahaina were outdated and did not incorporate advanced technologies such as IoT sensors or digital twins. Hawaiian Electric could not confirm downed power lines.
Redundancy and Network Reliability	X			Lahaina lacked redundancy in communication networks. When cellular lines went down, emergency response teams used closed channel radios that public officials could not listen to.
Information Sharing	X			Despite an “unprecedented advance warning” from forecasters, minimal messages were relayed to the community. Police and fire crews delivered warnings

				door to door. Emergency sirens were never used.
Cybersecurity	X			Although specific instances of cybersecurity attacks were not present during these fires, little emphasis was placed on cyber-threats due to the overwhelming failure of physical failures.
Scalability	X			The communication breakdowns during the wildfires left authorities in the dark and residents without emergency information, demonstrating the system's inability to scale and adapt during the crisis

5.1.2 Requests for Change

The following table describes specific changes to current methods justifying them for the target digital twin architecture. Most justifications revolved around the need for real-time data and communication focusing on proactive preparation instead of reactivity.

Table #5: Request for Change

Criteria	Issue	Change Proposal	Justification
Data-driven decision making	Lack of real-time data hindered educated decision making and proactive measures.	Integrate sensor data from a multitude of sources (weather conditions, fire spread models, infrastructure status)	Real-time data is required to gain situational awareness and make informed decisions
Integration of Technology	Limited use of technology to enhance situational awareness and decision making	Develop a centralized command platform that integrates emergency alert systems, cellular networks, and IoT-based smart city sensors. This could also simulate various disaster scenarios and test how different communication methods perform under stress.	Having a central base allows for overhead views of the situation that on-site responders may not have access to. Furthermore, a single headquarters for digital twin management would allow for quick and easy updates/modifications.
Redundancy and Network Reliability	Lack of network redundancy led to significant	Model redundant communication networks that integrate satellite internet,	Without reliable networks,

	communication failures	cellular on wheels, and backup power systems. The Digital Twin could run stress tests to identify weak points and recommend optimal placement of backup communication nodes.	communication cannot occur.
Information Sharing	Lack of coordination of local, state, and federal agencies crippled response efforts and resulted in minimal information shared back to the public.	Create a shared emergency response framework that simulates multi-agency coordination and provides a single-source-of-truth platform.	Officials shared information to a Facebook page that reached a small number of the public. Moreover, when the responders switched to radio, public officials were no longer in the loop.
Cybersecurity	Although physical failures were the main concern, further network disruptions from cyber threats are a real concern.	Simulate cyberattack scenarios on the emergency communication network, allowing cities to test resilience against hacking, jamming, and misinformation campaigns. AI-based anomaly detection would help secure emergency communication channels	Although this change proposal has a lower priority than others, it should certainly be considered.
Scalability	Communication networks could not be scaled appropriately during failures.	Design a dynamic network model that can scale automatically by activating ad-hoc mobile networks (drones, emergency Wi-Fi stations, vehicle-mounted repeaters, THOR/COW) when fixed infrastructure fails.	The ability to scale communication networks during emergency events is crucial to maintaining network reliability.

5.1.3 Business Capability Map

In accordance with TOGAF and DT consortium standards, the 6 capability types aligned with DTs in the DT capability periodic table had specific elements selected or denied based on plausibility in the project. Subjects such as the gaming engine visualization were rejected as there was no room within the three-year program to adapt VR technology for video game-based fire training. Synthetic data generation from real-time data acquisition sources and IoT sensors were

selected for usage of acquiring real-time data and converting to synthetic data generation for stress testing and training future emergency responders. (See the appendix for more information.)

DS.AI Data Acquisition & Ingestion	DS.SG Synthetic Data Generation	IR.ET Enterprise System Integration	IC.SR Search	IC.PR Prediction		UX.BV Basic Visualization	UX.DB Dashboards
DS.ST Data Streaming	DS.ON Ontology Management	IR.EG Eng. System Integration	IC.CC Command & Control	IC.AI Artificial Intelligence		UX.AV Advanced Visualization	UX.CI Continuous Intelligence
DS.TR Data Transformation	DS.RP Digital Twin (DT) Model Repository	IR.IO OT/IoT System Integration	IC.OS Orchestration	IC.PS Prescriptive Recommendations		UX.RM Real-time Monitoring	UX.BI Business Intelligence
DS.CX Data Contextualization	DS.IR DT Instance Repository	IR.DT Digital Twin Integration	IC.AL Alerts & Notifications	IC.FL Federated Learning	IC.BR Business Rules	UX.ER Entity Relationship Visualization	UX.BP BPM & Workflow
DS.BP Batch Processing	DS.DS Domain Specific Data Management	IR.CL Collab Platform Integration	IC.RP Reporting	IC.SM Simulation	IC.DL Distributed Ledger & Smart Contracts	UX.XR Extended Reality (AV/VR/MR)	UX.GE Gaming Engine Visualization
DS.RT Real-time Processing	DS.SA Data Storage & Archive Services	IR.AS API Services	IC.AA Data Analysis & Analytics	IC.MA Mathematical Analytics	IC.CS Composition	UX.GM Gamification	UX.3R 3D Rendering
DS.AS Asynchronous Integration	DS.SR Simulation Model Repository	MG.DM Device Management	MG.EL Event Logging	TW.EC Data Encryption	TW.SC Security	TW.SF Safety	TW.RP Responsibility
DS.AG Data Aggregation	DS.AR AI Model Repository	MG.SM System Monitoring	MG.DG Data Governance	TW.DS Device Security	TW.PR Privacy	TW.RL Reliability	TW.RS Resilience

Table #6: Business Capability Map

5.1.4 Target Data Architecture

After building the application, data, business function, process, and conceptual realization diagrams, the target data architecture was formed. This target architecture integrates real-time and historical data to support a resilient system for emergency response and communication infrastructure. Real-time inputs like emergency response efforts, power grid status, cellular network traffic, and fire/weather data are analyzed to assess communication content, network health, and environmental conditions. This information is then augmented with historical and geographical GIS data, cellular tower infrastructure, and emergency team locations. This data can be used to support actionable insights like public alerts, inter-agency coordination, alternative network suggestions, and network scaling. Additionally, this digital twin could be used to stress-test communication networks to identify weaknesses before disaster strikes. All in all, this enhanced target architecture will allow for data-driven decision-making while effectively reducing cognitive load and increasing situational awareness.

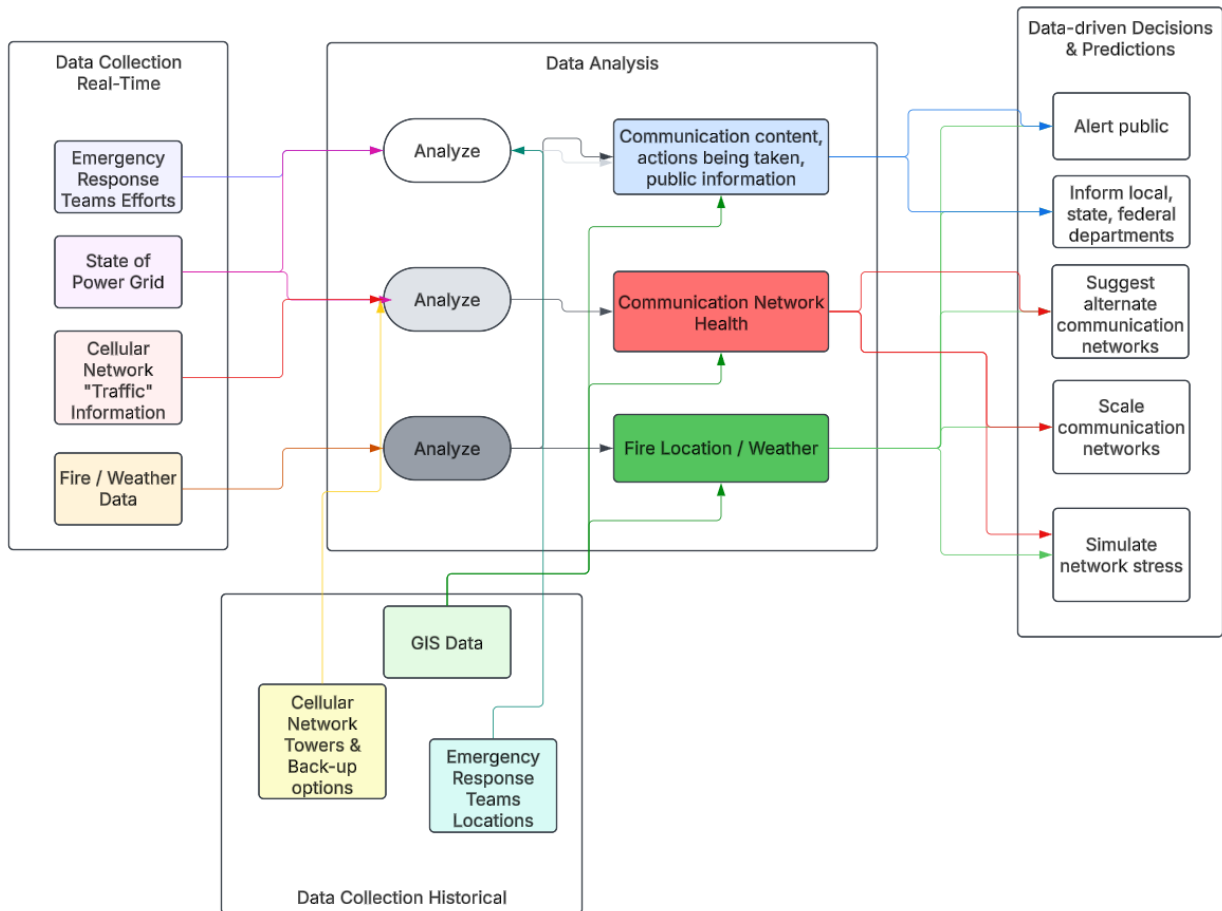
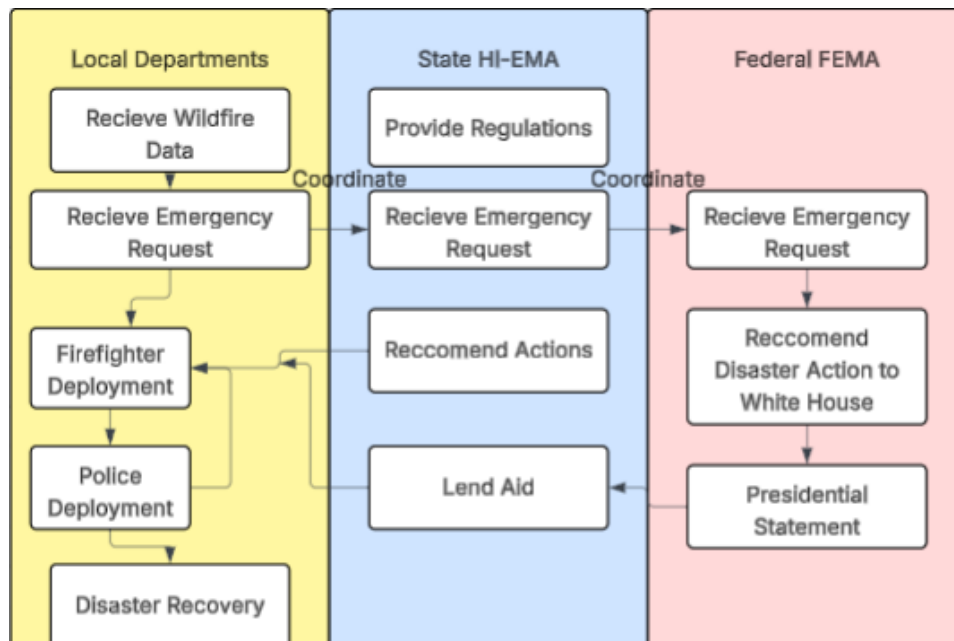


Figure #1: Target Data Architecture

5.2 Baseline Business Architecture



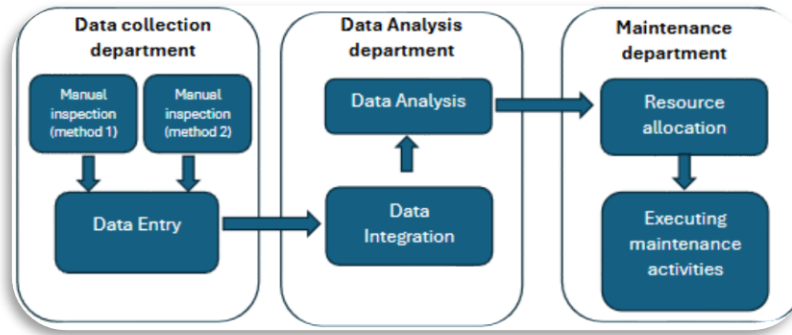


Figure #2: Baseline Business Architecture

The baseline business architecture demonstrates the organizational requirements and need for the business. When a natural wildfire disaster occurs, there is a local response to urban incidents, but higher responses require emergency requests for aid to Hawaii emergency services (HI-EMA) and the federal level (FEMA) for aid. Regulations provided by the state are followed at the local level with recommended actions primarily occurring in Hawaii state and lower level. The goal is for efficient and effective response strategies hinging on the communication between these departments and their responsibilities at each geographical level.

5.2.1 BPMN Diagram

A BPMN diagram is a visual representation of the business process. Here the wildfire emergency is related to the public, emergency worker agencies, power companies, and cell tower companies. The main starting points involve either a sensor reading a downed system from a company or an alert to 911 from the public. Once the fire risk arrives it is assessed as an emergency event or not. Given it is an emergency event, the emergency workers respond accordingly to the fire sending firefighters or electricians to fix networks. Given consideration of the state of emergency, city officials send out an emergency warning to the public and act accordingly with the fire and police department following evacuation if necessary.

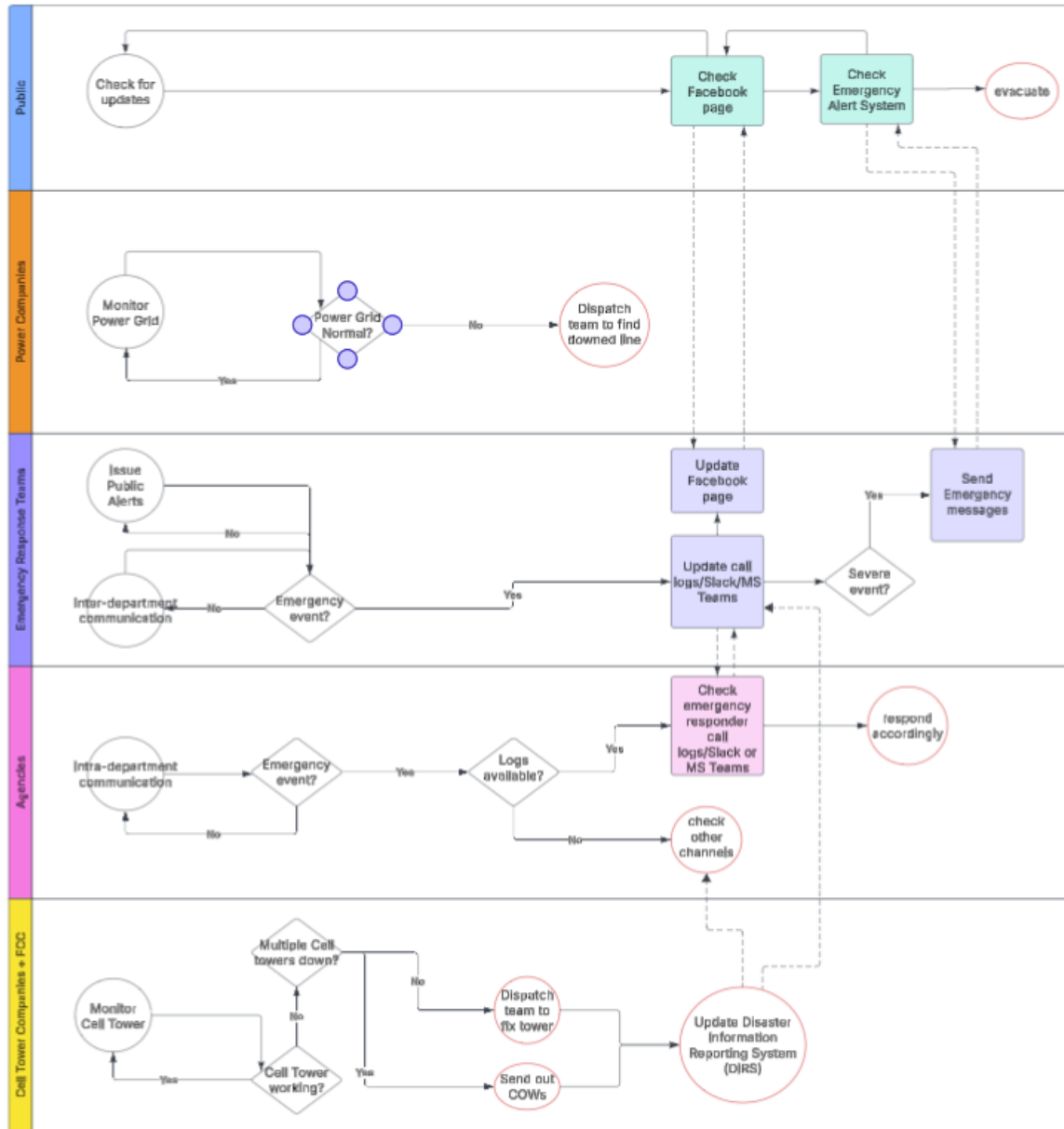


Figure #3: BPMN Diagram

5.3 Technology Architecture

5.3.1 Application/Technology Matrix

The following table showcases how each relevant technology will affect the procedures and methods in each of the application groups. This table shows that a given technology may be applicable to not just the main application group it is intended for.

Application Groups	Technology Groups	Cell Towers	Power Grid	MS Teams	Custom API Services	CLSS	Social Media	Modeling Software	Satellites	Sensors	IPAWS
Cell Towers											
Monitoring		++	-	+	++	+	+	-	-	+	-
COW Deployment		++	-	+	++	-	-	+	-	-	-
Power Grid											
Monitoring		-	++	+	+	+	+	-	-	+	-
Generator Deployment		-	++	+	-	-	-	+	-	-	-
Information Sharing											
Public Alerts		-	-	-	-	++	++	-	-	-	++
Update Responders/Officials		-	-	++	-	++	+	+	+	-	++
Event Recording		+	+	+	+	+	+	-	-	+	+
Fire Status											
Evacuation Rec's		+	-	+	-	++	++	++	+	-	++
Attack Plan		-	-	++	-	-	-	++	++	+	-

Figure #4: Application/Technology Matrix

6. Roadmap

A three-year roadmap was developed to define the projects required to implement NetProtect as the communications hub for natural disaster relations in Lahaina. The roadmap was separated into four distinct phases based on the specific challenges and identified improvements after the 2023 fires. These phases are sensor deployment and data integration, simulation and redundancy modeling, training data and predictive analysis, and community lifeline integration. The gap analysis identified a serious need for technological improvement with the city's wildfire sensors and cell towers as faster detection would allow for quicker responses. The primary task over the first year is to install sensors alongside the reconstruction of the city. In collaboration with the Department of Homeland Security (DHS) Science and Technology Directorate (S&T) over 64 stationery and 16 mobile wildfire detection sensors will be deployed including N5SHIELD/DRYAD SILVANET wildfire sensors, Bosch BME688 sensor, AVNET Silica cell traffic tower sensors, and TE power grid sensors (Chan, 2024). Working alongside construction crews, these sensors will monitor the local infrastructure energy and communication sources for potential fire-starting conditions, meteorological data, and downed networks. During this task, a map of Maui will be developed with 3D satellite geomapping to determine population density, geographical terrain and vegetation-based wildfire risk, and reevaluated evacuation routes. Once complete, this map will aid in determining risk factors for fire starting locations through the visible infrared imaging radiometer suite (VIIRS & MODIS) NASA near-real-time data satellite that will aid sensor placement (Johnson, 2008).

Additionally, enhanced knowledge of the residential areas after reconstruction will allow for superior evacuation routes which stalled traffic during the previous natural disaster. API services will be performed in conjunction with sensor installation creating API services for the cell towers to provide info on their traffic and health. Companies such as AT&T may provide some resistance to these efforts. Ideally, a real-time-data warehouse provided by Estuary will be utilized for immediate analysis of performance and customer usage patterns. While complex and costly with high operational costs, Estuary will provide faster data ingestion and processing speeds, immediate analysis of network performance, and even instantaneous data on grid performance and outage situations. The last project of the first phase involves developing a visualization platform with a web application dashboard that provides a map of key communication resources, emergency personnel, fire risk, and downed systems combining the year of effort into one system to monitor Lahaina's wildfire risk and communication network.

The second phase comprised of simulation and redundancy modeling utilizes computational models exploring system behavior in real-time while addressing reliability and resilience when handling failure. A digital model of Lahaina is created in semi-real time, focusing on on-site monitoring from previously mentioned sensors, social media fire reports, traffic mobility from sensors, and historical geography of communication networks at high risk. The aim is to virtually represent the state of Lahaina, population residences, and the surrounding area for risk simulation alongside their communication pathways. Geometric data of the area, historical data of 911 calls and wildfires, and behavioral data of fire reports with social media API scripts are all utilized to form a model simulation for communication inside of emergency departments, public communication, and the interactions between them within the natural disaster DT (Li, 2025). This model will allow for synthetic data to be generated allowing for testing of redundancy enabling proactive preparation. The DT intradepartmental communication network will set up a communication network app with the DT to aggregate, identify, and send data for each of the departments and emergency service personnel over the course of the second year. Software and computer engineers will update system technology across all departments and systems for greater interoperability within the DT communication network. WatchDuty is an app already expediting this process providing wildfire mapping and alert app powered by real people giving real-time information vetted by trained professionals. Integrating this app with the DT network will allow enhanced interaction between the DT and emergency services in real-time to

locate and communicate wildfire information between themselves and the public with an enhanced emergency response system.

The third phase focuses on machine learning training as a project to enhance data and predictive analysis given risk factors. DTs allow for predictive analysis enabling users to activate early warning systems and estimate the potential of disasters with enhanced organization of data volume, variety, and velocity. This information when transferred can allow for disaster management strategies to optimize resource allocation and minimize harmful impacts from safety to monetary damage. A data-driven model similar to other wildfire DT monitors can leverage vast amounts of data to apply more accurate predictions (Huang, 2024). Historical and real-time sensor wildfire data such as weather reports, temperature, wind speed, humidity, 911 calls & social media disaster reports can be used as inputs with synchronous interactions with the virtual AI wildfire prediction and communication stability model. Synthetic training data can be generated for fire behavior, spreading rate, and downed networks. Training simulations for first responders will be formed with VR fire science & immersion fire training allowing for enhanced decision-making capabilities such as when to utilize wireless protective services between officers or Skynet to communicate danger to the public.

The final phase involves the integration of this digital twin with the community as a lifeline. Final installations of API services to monitor social media for fire reports, WatchDuty downloads by the public, and the Community Lifeline Status System (CLSS) to be adapted by emergency workers occur during this phase. The primary method for fire reporting comes from 911 calls; therefore, utilization of social media, Slack, and Watch Duty with #wildfire to signify current events of natural disasters will greatly enhance relaying information to the community from the DT. The CLSS operationalizes real data for real-time impact assessments, highlighting critical community impacts and enhancing your community's resilience (Rojahn, 2017). This system provides greater situational awareness with data-informed reporting. It manages and analyzes dynamic geospatial data while registering existing data and links to critical community impact indicators. The final failure of the 2023 fire was the lack of emergency messaging through IPAWS to the public, with this community lifeline the final step of transferring information back and forth from the public to officials to the digital twin can be complete.

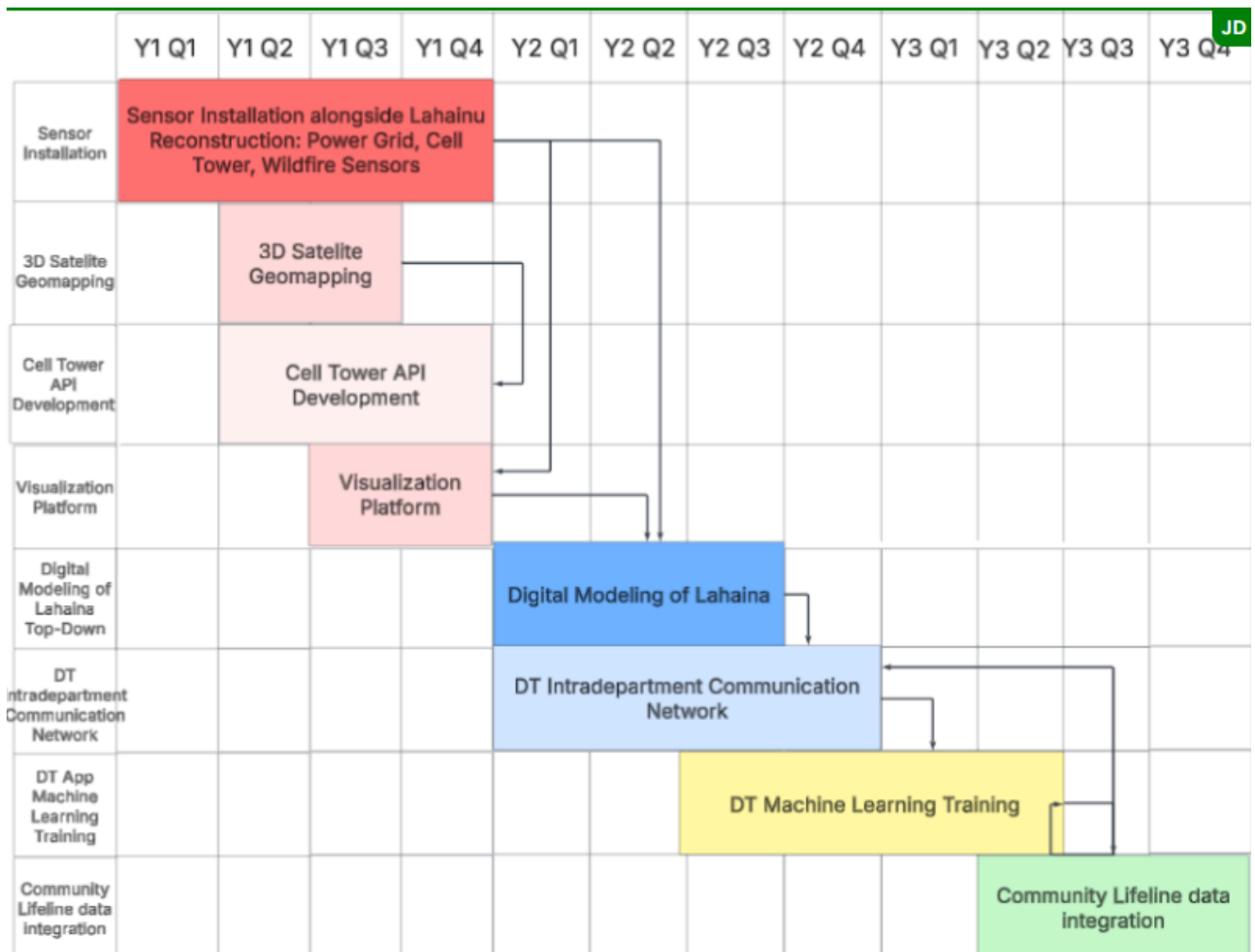


Figure #5: NetProtect Roadmap

7. Conclusion

Given the wildfire disaster in Lahaina in 2023, this project established a three-year plan to develop a DT for communications during natural disasters. The lack of adequate sensors for real-time monitoring of the wildfire and infrastructure alongside failures in communication halting evacuation are addressed in this model. A widespread network of sensors and API technology allow for NetProtect to observe real-time wildfire data and communication risks integrating communication from governmental departments to the public in a model of Lahaina. Predictive analysis, an app visualization platform, and a training system for emergency personnel will improve the response of the community to wildfires. This establishes the goal of forming a communication network that will keep the people and land of Lahaina safe.

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9. Appendix - A

Table# 6: Business Capability Table

	Name: Data Acquisition and Ingestion	Type: Data services	Classification: Basic
Ability:	The ability to configure and acquire data from different data sources including control systems, historians, IoT sensors, smart devices, engineering systems, enterprise systems etc.		
Purpose:	The purpose is to acquire data from the physical world, engineering technology systems, and information technology systems to support subsequent processing and insight generation.		

	Name: Data streaming	Type: Data services	Classification: Basic
Ability:	The ability to transfer large volumes of data continuously and incrementally between a source and a destination without having to access all data at the same time.		
Purpose:	The purpose is to acquire fast continuous packets of information which are changing at high speed to be able to get near real-time insights.		
	Name: Data Transformation	Type: Data services	Classification: Basic
Ability:	The ability to convert data types and properties through cleaning, structuring and enriching raw data to make it suitable for further processing and analytics		
Purpose:	The purpose is to make data useable in Digital Twins		
	Name: Real-time Processing	Type: Data services	Classification: Basic
Ability:	The ability to manage and act on the captured data with minimal latency		
Purpose:	The purpose is to support immediate insights from the data.		
	Name: Data Aggregation	Type: Data services	Classification: Basic
Ability:	The ability to gather raw data and express in a summary form		
Purpose:	The purpose is to gather data from multiple sources with the intent of combining these data sources into a summary for data analysis		
	Name: Synthetic Data Generation	Type: Data services	Classification: Specialist
Ability:	The ability to generate synthetic data based on patterns and rules in existing sources		
Purpose:	The purpose is to create representative synthetic data that can be used by the digital twin to train and score predictive models		
	Name: Temporal Data Storage	Type: Data services	Classification: Basic
Ability:	The ability to store, organize and retrieve data relating to time instances through temporal data types, and stores information relating to past, present and potentially future time.		
Purpose:	The purpose is to store, manage and retrieve temporal (timeseries) data		
	Name: Data Storage & archive services	Type: Data services	Classification: Basic
Ability:	The ability to store, organize and retrieve data based on how frequently it will be accessed and how long it will be retained.		

Purpose:	The purpose is to reduce the cost and effort of managing Digital Twin data by using hot, cold and archival data services		
	Name: Simulation Model Repository	Type: Data services	Classification: Enhanced
Ability:	The ability to store, manage and retrieve algorithmic codebases, business rules and meta data that describe a simulation model.		
Purpose:	The purpose is to register and manage a portfolio of simulation models in a central repository to improve configuration management and model governance.		
	Name: AI Model Repository	Type: Data services	Classification: Basic
Ability:	The ability to store, manage, search and retrieve the algorithmic codebase that describe an artificial intelligence (AI) model or machine learning (ML) model		
Purpose:	The purpose is to register and manage a portfolio of AI and machine learning models in a central repository to improve configuration management and model governance.		
	Name: Collab Platform Integration	Type: Integration	Classification: Basic
Ability:	The ability for the digital twin to interface with platforms like Yammer, Jabber, Teams, Slack		
Purpose:	The purpose is to integrate collaboration platforms to provide Digital Twin users with a conversational user interface		
	Name: API Services	Type: Integration	Classification: Basic
Ability:	The ability for the digital twin to publish APIs to external, partner, and internal developers to access data and services.		
Purpose:	The purpose is to simplify Digital Twin development by allowing Digital to integrate to products and services without knowing how they are implemented		
	Name: OT/IoT System Integration	Type: Integration	Classification: Basic
Ability:	The ability to integrate directly with control systems and IOT devices/sensors, SCADA		
Purpose:	The purpose is to integrate operation technology (OT) and IoT applications that data to flow between Digital Twin systems with ease		
	Name: Edge AI and Intelligence	Type: Intelligence	Classification: Basic

Ability:	The ability to make decisions at the device level based on real -time data, distribution and federation of analytics at the edge instead of transporting the data to the cloud to perform analytics.		
Purpose:	The purpose is to make real-time decisions near field.		
	Name: Alerts and Notifications	Type: Intelligence	Classification: Basic
Ability:	The ability to display and manage alerts, messages, message queues, triggers, and notifications		
Purpose:	The purpose is to trigger actions which may require intervention in the ongoing processes.		
	Name: Reporting	Type: Intelligence	Classification: Basic
Ability:	The ability to generate configurable and customizable reports to get insights into the data		
Purpose:	The purpose is to get insights into the data which can be useful for various stakeholders in the system as well as for regulatory compliance.		
	Name: Data Analysis and Analytics	Type: Intelligence	Classification: Basic
Ability:	<i>The study and presentation of data to create information and knowledge.</i> The ability to analyze data through charts, tables, dashboards, fetch data between dates, and filter data based on various criteria. The analysis of data, typically large sets of business data, using mathematics, statistics, and computer software with an objective to draw conclusions.		
Purpose:	The purpose is to understand past trends from historical data.		
	Name: Prediction	Type: Intelligence	Classification: Basic
Ability:	The ability to estimate that a specified event will happen in the future or will be a consequence of other events.		
Purpose:	The purpose is to use historical data, engineering, and analytical models to predict future events before they occur.		
	Name: Machine Learning	Type: Intelligence	Classification: Basic
Ability:	The ability of computer algorithms that improve a digital twin automatically through experience. The algorithms build a mathematical model based on "training data", to make predictions or decisions without being explicitly programmed to do so. It is seen as a subset of artificial intelligence.		
Purpose:	The purpose is to enable the digital twin and digital twin systems to learn from data, identify patterns, and make decisions with minimal human intervention.		

	Name: Artificial Intelligence	Type: Intelligence	Classification: Basic
Ability:	The ability for a system to perform actions and take decisions like humans. AI would include machine learning, natural language processing, knowledge modelling and representation, reasoning, inferencing etc. It is based on the capacity of a computer to perform operations analogous to learning and decision making in humans, as by an expert system, a program for CAD or CAM, or a program for the perception and recognition of shapes in computer vision systems.		
Purpose:	The purpose is to enable a digital twin or a digital twin system to take actions and decisions similar to humans.		
	Name: Simulation	Type: Intelligence	Classification: Specialist
Ability:	The ability to create approximate imitation of a process or a system using past historical information, physical models, video, audio, and animation, what-if-scenarios.		
Purpose:	The purpose is to imitate the behavior of a physical system in the digital twin before applying it to the physical world. Training operations and maintenance teams on simulated digital twins is another purpose of simulation		
	Name: Prescriptive Recommendations	Type: Intelligence	Classification: Enhanced
Ability:	The ability to create prescriptive recommendations based on business rules and AI logic to suggest the best next actions to take when a pre-determined event happens.		
Purpose:	The purpose is to enable Digital Twins to provide guidance based on a combination of analytics, business rules and workflow to create actions and deliver business outcomes		
	Name: Advanced Visualization	Type: UX	Classification: Basic
Ability:	The ability to graphically or parametrically (that is, through parameters and values), visualize data through complex charts and graphs, dashboards fetching raw and process data from multiple systems, complex 3D models and animations, visualizations with overlayed data from different systems.		
Purpose:	The purpose is to help people understand the significance of data by placing it in a visual context.		
	Name: Real-time monitoring	Type: UX	Classification: Basic
Ability:	The ability to present and interact with continuously updated information streaming at zero or low latency.		
Purpose:	The purpose is to help make decisions which are of consequence to real-time.		
	Name:	Type:	Classification:

	Entity Relationship Visualization	UX	Basic
Ability:	The ability to present Digital Twin entities and their hierarchical or graph-based relationships in an interactive way		
Purpose:	The purpose is to help business users navigate and interact with complex entity (asset) hierarchies in a user-friendly manner		
	Name: Dashboards	Type: UX	Classification: Basic
Ability:	The ability to provide a graphical user interface which provides at-a-glance views of key performance indicators relevant to a particular objective or business process.		
Purpose:	The purpose is to enable various personas in operations, technology, and business to visually understand the current or past state of a system.		
	Name: Continuous Intelligence	Type: UX	Classification: Basic
Ability:	The ability to analyze data in flight (signals) to derive insights and actions in a business user focused visual interface		
Purpose:	The purpose is to have various personas in operations, technology, and business to make informed real-time decisions.		
	Name: BPM & workflow	Type: UX	Classification: Basic
Ability:	The ability to execute a sequence of actions as a process flow to achieve specific business outcomes		
Purpose:	The purpose is to have effective, repeatable actions that deliver the business outcomes of the Digital Twin		
	Name: Device Management	Type: Management	Classification: Basic
Ability:	The ability to provision and authenticate, configure, maintain, monitor and diagnose connected IoT devices operating as part of Digital Twin environment		
Purpose:	The purpose of (IoT) device management is to provide and support the whole spectrum of functional capabilities of the devices and sensors		
	Name: Data Governance	Type: Management	Classification: Basic
Ability:	The ability to manage the availability, usability, integrity and security of the data in Digital Twin systems, based on internal data standards and policies that also control data usage.		
Purpose:	The purpose is to ensure that data is consistent and trustworthy and doesn't get misused.		
	Name: Device security	Type: Trustworthiness	Classification: Basic

Ability:	The ability to enforce authenticated and authorized access to IoT device data through identity management, role-based access, encryption, and policies		
Purpose:	The purpose is to control access to device data by having the appropriate privileges and enforcement framework for users and programs.		
	Name: Security	Type: Trustworthiness	Classification: Basic
Ability:	The ability to protect Digital Twins from unintended or unauthorized access, change or destruction. Security concerns equipment, systems and information, ensuring availability, integrity and confidentiality of information.		
Purpose:	The purpose is to control access to digital twin data by having the appropriate privileges and enforcement framework.		
	Name: Safety	Type: Trustworthiness	Classification: Basic
Ability:	The ability to operate digital twins without causing unacceptable risk of physical injury or damage to the health of people, either directly, or indirectly because of damage to property or to the environment		
Purpose:	The purpose is to ensure the digital twin and insights it provides to not cause harm.		
	Name: Reliability	Type: Trustworthiness	Classification: Basic
Ability:	The ability of a Digital Twin system or component to perform its required functions under stated conditions for a specified period of time. This includes expected levels of performance, QoS, functional availability and accuracy.		
Purpose:	The purpose is to ensure the information provided to the digital twin and by the digital twin is consistently available.		
	Name: Resilience	Type: Trustworthiness	Classification: Basic
Ability:	The ability of a Digital Twin system or component to maintain an acceptable level of service in the face of disruption. This includes the ability to recover lost capacity in a timely manner (using a more or less automated procedure), or to reassign workloads and functions.		
Purpose:	The purpose is to ensure the digital twin can respond appropriately given the event of a failure.		